

Immunostaining of fibrotic tissue isolated from the vitreoretinal interface

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Background: Fibrotic membrane can form at the vitreoretinal interface e.g. after the healing of a surgical wound or as a result of a disease such as rhegmatogenous retinal detachment, proliferative vitreoretinopathy, age-related macular degeneration, or various pediatric diseases ¹⁻⁵. Fibrotic membrane consists of cells that have undergone epithelial-mesenchymal transition (EMT), proliferated, and migrated, e.g. RPE cells, glial cells, fibroblasts and myofibroblasts, as well as excessive extracellular matrix (ECM) ^{1,2}. For example, vimentin and α -SMA are key indicators of the transition of mesenchymal cells into myofibroblast, which promotes the formation of fibrosis ⁶. Fibrosis leads to wrinkling, folding, contraction, and detachment of the retina ².

Methods: Vitreoretinal interface scar tissue was placed in a well of 12-well plate and photographed using a light microscope (Zeiss Axio Vert). The membrane was fixed with 4% PFA, washed with DPBS, blocked with 1% BSA / 0.1% Triton X-100 in DPBS, and washed again with DPBS. The membrane was stained with primary antibodies to α -SMA and vimentin and then treated with Alexa Fluor secondary anti-mouse and anti-rabbit antibodies. Nuclei were stained with Hoechst 33342. Membrane was washed, mounted with Mowiol, and photographed using Olympus Apex View APX100.

Results: In the photograph, the membrane appeared thin, light, and almost translucent. Hoechst staining revealed that the membrane contained occasional cells and cell clusters, as well as abundant ECM. The stained scar tissue contained myofibroblast markers α -SMA and vimentin.

Conclusions: α -SMA and vimentin appeared to be at least partially localized to the nuclei staining in scar membrane. The role and inhibition of α -SMA and vimentin in membrane formation require further studies. Nuclei staining revealed occasional cells in the membrane, so the role and inhibition of cell migration and ECM production also require further studies.

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References

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